

to Useful Systems'. It was started as a research project in the computer science department of the University of California, Santa Barbara. Though this product has been commercialised, the open source version is still maintained. The current release is Eucalyptus 4.0. It is the ideal choice for hosting IaaS due to its ease of use and AWS API compatibility. Eucalyptus consists of six distinct components, which are grouped into three separate levels, i.e., the cloud level, the cluster level and node level. All these components interact with each other using SOAP messaging with WS-security.

There are two components in the cloud level: cloud controller (CLC) and scalable object storage (SOS). CLC is responsible for authentication, accounting, reporting and quota management. SOS resembles AWS S3 (Simple Storage Service) and provides basic storage implementation known as Walrus. The cluster level consists of three components. The cluster controller (CC) is the front-end for the cluster within a Eucalyptus cloud and it communicates with other components. The CC manages virtual machine execution and SLAs set per cluster. The storage controller (SC) equivalent to AWS is elastic block store (EBS). SC is responsible for providing the block storage to virtual machines, and it interfaces with storage systems including local, NFS, iSCSI and SAN. The third component in the cluster level is VMware broker and this is an optional component. VMware broker mediates interactions between the CC and ESX/ESXi/vcentre servers. The last level, i.e., the node level, consists of the node controller (NC), which is responsible for hosting virtual machine instances and actual workhorses for the cloud in providing IaaS. Refer to eucalyptus.com for more information.

CloudStack

CloudStack is open source software designed to deploy and manage large networks of virtual machines, as a highly available, highly scalable IaaS cloud computing platform. CloudStack development was started by *Cloud.com*. Later, in 2010, *Cloud.com* released most of CloudStack as free software under GNU GPLv3. Citrix purchased *Cloud.com* in 2011 and, later, Citrix donated CloudStack to the Apache Software Foundation (ASF). CloudStack supports most of the hypervisors: VMware ESXi, KVM, XenServer, Oracle VM and Hyper-V. It provides a rich, intuitive Web based user interface, command line utilities and full-featured RESTful APIs for managing the compute, networking, storage, and software entities in the cloud. The current stable release, Apache CloudStack 4.4.2, allows users to build feature-rich public and private cloud environments.

The key features of CloudStack are:

- One cloud, multiple hypervisors
- Massively scalable infrastructure management
- Easy-to-use Web interface
- Robust RESTful APIs

The CloudStack implementation consists of two major parts: the management server that contains the core logic,

and the server resources that contain the translation layer to interact with the hardware. The management server manages the resources provisioned within the cloud such as the hosts, storage devices and IP address. In the CloudStack cloud, typically, there will be one or more zones managed by each management server.

CloudStack has the hierarchy of *Zone --> Pod --> Cluster --> Host*, and comes with primary and secondary storage. A zone is equivalent to a single data centre, consisting of one or more pods and secondary storage. A pod usually represents one rack of hardware that includes a Layer-2 switch and one or more clusters. A cluster consists of one or more hosts installed with a hypervisor and a primary storage that provides block storage. Primary storage is associated with a cluster and provides the disk volumes for VMs running in that cluster. Secondary storage is associated with zones and hosts templates, ISO images and disk volume snapshots. In addition, CloudStack has a few system-VMs that take care of most of the core logic, i.e., the storage and networking pieces.

You can refer to cloudstack.apache.org for further reading.

Advantages of IaaS and open source cloud platforms

IaaS, as you can infer, is a perfect service offering if you are looking for the following features:

- No investment in hardware
- Scalability
- Flexible style costs
- Physical security of data centre locations
- No single point of failure

Infrastructure as a Service is a form of cloud computing that provides the user with computer services over the Internet by the use of hypervisors and virtual machines. IaaS is one of the three main cloud computing services, the others being Software as a Service (SaaS) and Platform as a Service (PaaS). IaaS services are hosted by third party providers, and the end users are billed for services on a pay-as-you-use basis. An important aspect of IaaS services is that the end user need not worry about administrative tasks such as dynamic scaling, software updates, desktop virtualisation, etc. The pay-per-use model is ideal for effecting huge cost savings. A word of caution though—users should check their bills carefully to avoid paying for services not availed of. 

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